

Project Proposal: Deep Neural Network Compression & CV

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Objective

This project guides students through a full computer vision pipeline, from image processing and ML fundamentals to deep learning (CNNs) and advanced model optimization. The goal is to build intuition for efficiently deploying large-scale neural networks on resource-constrained devices, such as on-device AI for **mobile phones** and **UAVs (drones)**.

Final Project Focus

The final project focuses on building a 3-stage compression pipeline to drastically reduce the storage footprint of the **AlexNet** model trained on the **CIFAR-10** dataset. By designing and combining **L1 pruning**, **K-Means quantization**, and **Huffman coding**, the goal is to achieve a highly efficient model. The target is an overall $9\times$ **compression ratio**, reducing the model from **228 MB** to **~ 25 MB** with a minimal and acceptable accuracy loss of approximately 1.5%.

Work to be Done

1. Foundations of Computer Vision

- Understand fundamental image representations (RGB, HSV color spaces) and basic image processing with filters (Gaussian Blur, Convolution).
- Learn frequency domain analysis (DFT, FFT) and feature extraction techniques like Edge Detection (e.g., Canny) and Hough Transforms.
- Implement data augmentation and preprocessing pipelines for CNNs, including random cropping, image resizing, and channel duplication as used for AlexNet.

2. Foundations of Machine Learning

- Understand the core concepts of supervised learning, differentiating between Regression (predicting values) and Classification (predicting labels).
- Implementing Linear Regression & K-Means to have hands on experience in Python, which will serve as the foundation for model weight quantization.
- Learn the core concepts of PyTorch: understand **tensors**, autograd, and key packages like `torch.nn` and `torchvision`.

3. Intro to Deep Learning and CNNs

- Build a basic Artificial Neural Network (ANN) using dense (fully-connected) layers and understand the fundamentals of backpropagation.
- Explore Convolutional Neural Network (CNN) architectures, learning the function of convolutional layers, pooling layers, and hierarchical feature maps.
- Implement and train the full AlexNet model on the CIFAR-10 dataset to establish a baseline for accuracy (e.g., $\sim 85-87\%$) and model size (228 MB).

4. Huffman Coding and Data Structures

- Review fundamental data structures, specifically binary trees and priority queues, which are prerequisites for the Huffman algorithm.
- Study the greedy algorithm behind Huffman coding for creating optimal, prefix-free codes for lossless data compression.
- Design and implement a custom Huffman coding scheme to encode quantized model weights, aiming to reduce the average bits per weight from a standard 5-bit encoding to **~ 3.57 bits**.

5. Pruning and Quantization

- Learn and apply **L1 magnitude-based unstructured pruning** to enforce sparsity in the network, aiming to remove 90% of the model's weights.
- Apply the K-Means algorithm (from part 2) to cluster the remaining non-zero weights, a technique known as weight quantization.
- Implement the quantization pipeline to compress the model's weights into a small number of clusters (e.g., **32 clusters**) to represent the entire weight distribution.

6. AlexNet Compression Model (Final Project Model)

- Integrate the 3-stage pipeline: Apply L1 pruning, then K-Means quantization, and finally Huffman coding to the baseline AlexNet model.
- Analyze the compression-accuracy trade-off, evaluating the final model's performance (target: **<1.5% accuracy loss**) against the baseline.
- Package the final compressed model and present a comparison of its metrics (e.g., **~25 MB**, 3.57 avg bits/weight) against the original (228 MB, 32-bit float) to validate the **9× compression** achievement.